

Light and Spectroscopy: Introduction

PURPOSE

In this Laboratory Experiment, we study the emissions spectra of an unknown gas. We will use the measured spectra to identify the gas by comparing it to a set of known emission spectra. As an example of a continuous spectrum, we will observe the spectrum of a light bulb and measure some wavelengths of visible light.

EQUIPMENT AND SUPPLIES

Spectrometer, gas lamp with mystery gas, diffraction grating, desk lamp with a light bulb, reference image with spectra of various gases and the Sun.

INTRODUCTION

The gas tube is filled with a single gas, which will produce a distinct emission spectrum. The light, that we observe from the lamp is a combination of all these emission lines. In order to measure the wavelength of a particular emission line, the incident light needs to be split into single wavelengths. One way to split light into single wavelengths is to use a diffraction grating.

Diffraction

A diffraction grating is a number of parallel slits that are very narrowly spaced. Due to the wave nature of light when waves of light illuminate a diffraction grating, light can bend around the edges, giving rise to bright and dark fringes, called a diffraction pattern. This bending of light around edges and small objects is called diffraction. Light of different wavelengths will bend differently, such that the bright fringes appear in different positions, thus spreading out the different parts of the light.

To see how a diffraction pattern arises, parallel rays fall on the diffraction grating as shown in the figure below. The angle θ , under which a bright fringe is observed, is proportional to the wavelength and proportional to the slit width, D .

Find the wavelength of a spectral emission line

By measuring the angle between the bright regions and the direction from the light source and by knowing the number of grating lines per unit length, one can determine the wavelength of the light that is producing the observed patterns of light. These quantities are related by the equation:

$$\lambda = D \times \sin \theta,$$

where λ (lambda) is the wavelength and θ (theta) is the angle between the direction to the light source and the direction to the image and D is the width of the slit.

Spectrometer

In this experiment, the diffraction angles will be measured using an instrument called a spectrometer. The spectrometer consists of an entrance slit, a collimator, a diffraction grating and a movable telescope. A collimator is a long tube which aligns the light into "columns" so that it hits perpendicular to the diffraction grating. The light entering the entrance slit is collimated and illuminates the grating. As shown in the figure below, the various spectral lines can be observed by moving the telescope from its center position to the left and right.

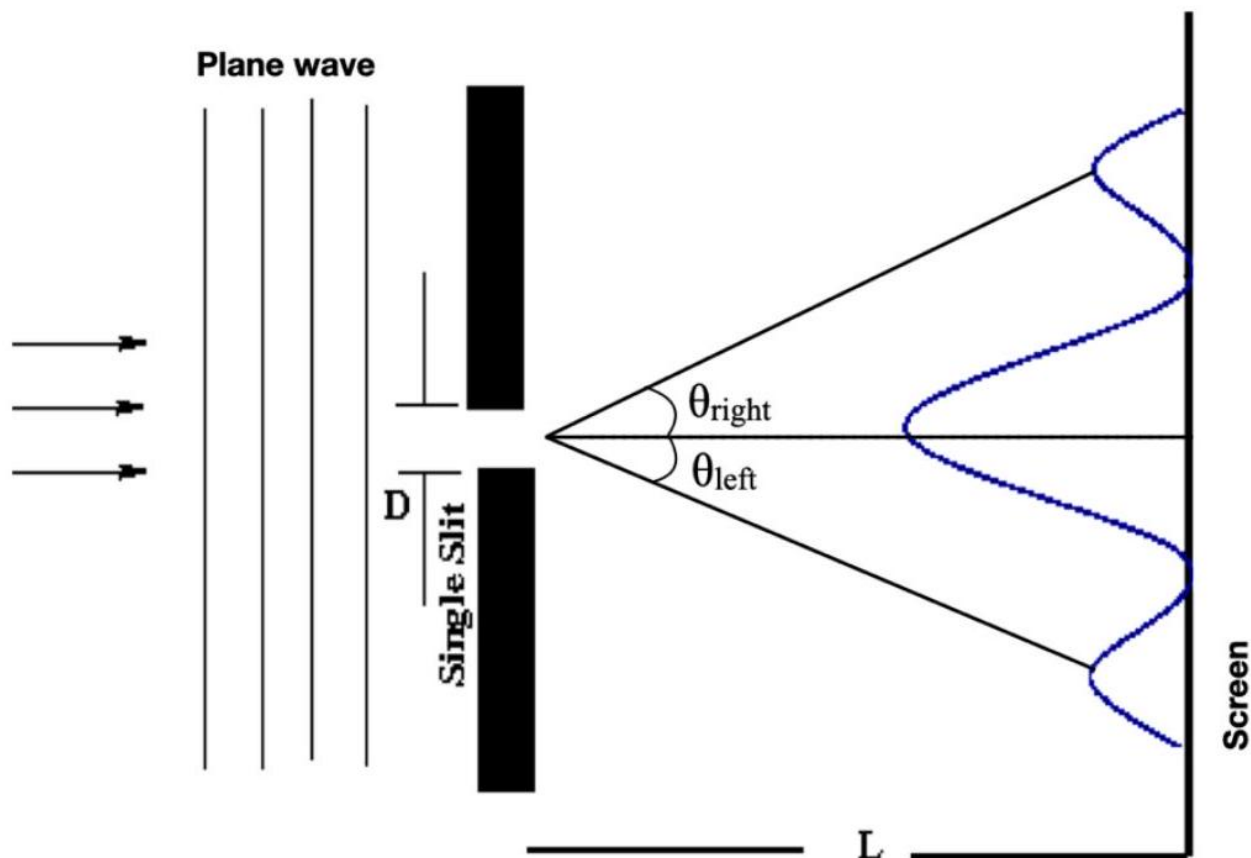


Fig. 1 Diffraction

Measuring the angle θ

The angle θ is calculated from the angular positions to the right (θ_{right}) and to the left (θ_{left}) of the center for the same spectral line (same color). The angular positions are read out from the Vernier.

Light and Spectroscopy: Procedure

I. Preparation

1. DO NOT MAKE ANY ADJUSTMENTS TO THE SPECTROMETER.
2. Turn on the gas lamp. In the warm-up period, observe the spectrometer setup.
 - a) The lamp illuminates the entrance slit.
 - b) The collimator and telescope are in a straight line.
 - c) The diffraction grating on the table of the spectrometer is perpendicular to the incident beam of light from the collimator, i.e. the front of the collimator and the grating are parallel.
 - d) Observe the image of the collimator slit through the telescope. You will see a bright, thin light in the telescope and a cross-hair. (You might have to move the telescope a little to the left or right in order to line it up with the cross-hair.)
 - e) If the image of the slit and cross-hair are not sharp, the instrument adjustments have been disturbed. Report to the instructor.
3. Record the number of lines per millimeter for your grating and the slit width, D , in nanometers.

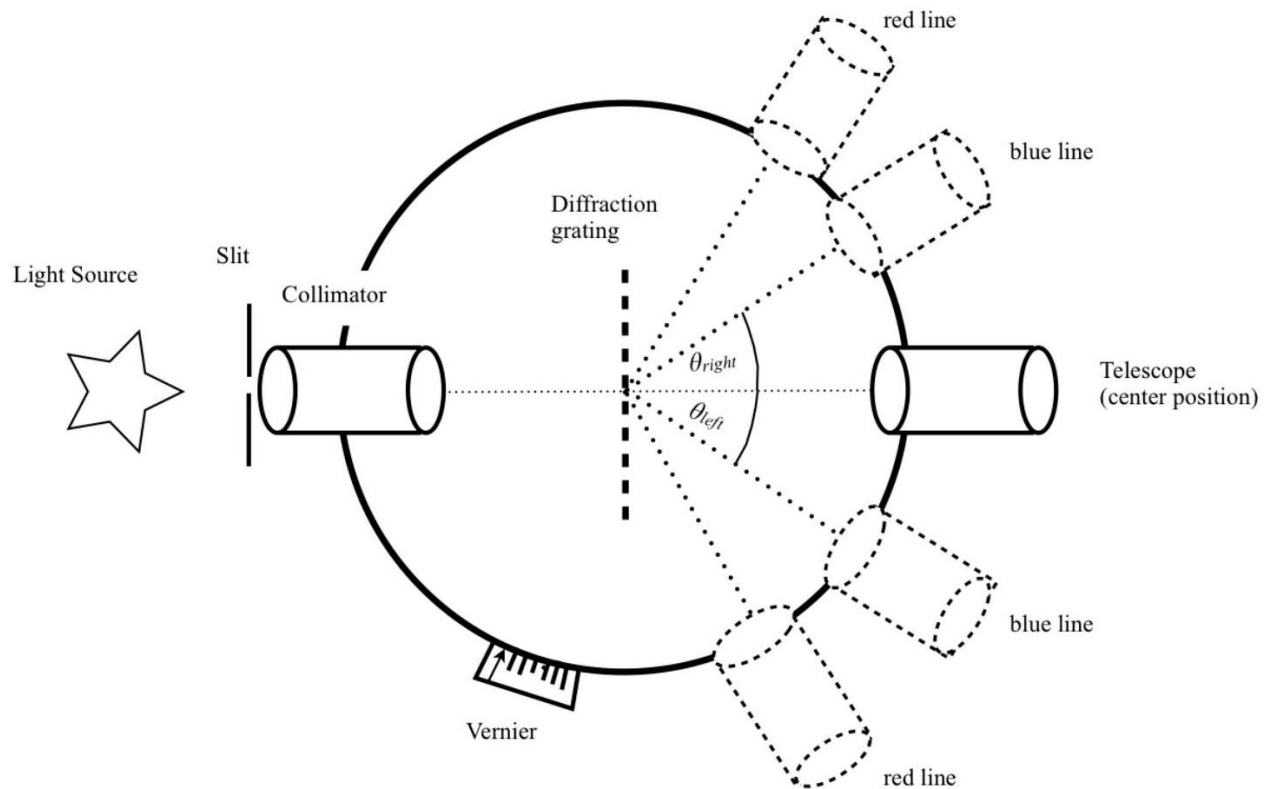
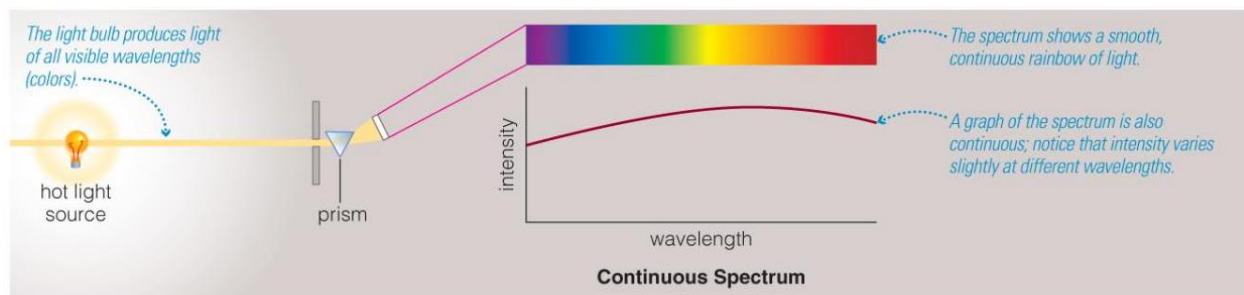


Fig. 2 Spectrometer

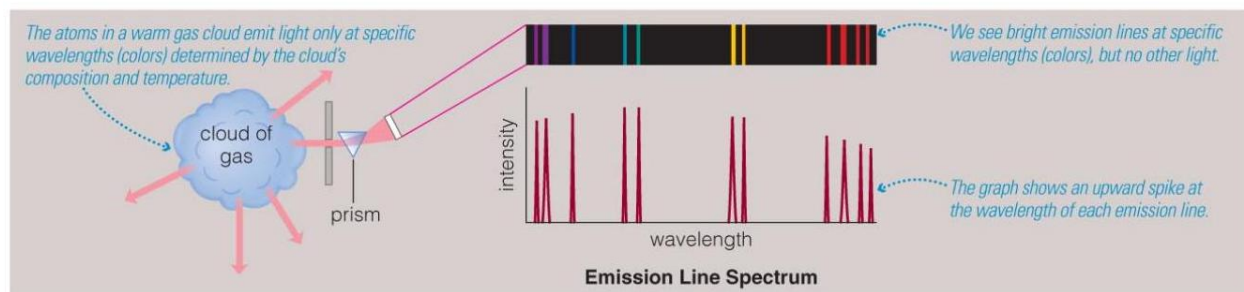
II. Measurement

1. Measure the diffraction angles for the emission lines
 - a) Swing the telescope to the right and the left sides from the central position (straight line up with collimator.) while looking through the telescope to line up the cross-hair with the emission line.
 - b) Do not record anything for the central blue line. That is the non-diffracted light from the lamp.
 - c) On your answer sheet, record the angular positions of the emission lines to the left (θ_{left}) and to the right (θ_{right}).
 - d) Calculate the difference in angles (what is the angle between the left and right positions) on your answer sheet. The diffraction angle, θ , is half the angular difference.
2. Using your measurement of the diffraction angle, calculate the wavelength of lines.
3. Repeat step 3 and 4 for the other colors/wavelengths.
4. On your answer sheet, use the colored pencils to make a graph of emission lines at the measured wavelengths.
5. Change your light source to the desk lamp with the lightbulb.
6. Look through the grating at the slit illuminated by "white" light produced by the lightbulb.
 - a) Select a particular blue region in the spectrum in the right image and record the angular position.
 - b) Then measure the angular position of the blue in the left image.
7. Using your measurement of the diffraction angle, calculate the wavelength of region.
8. Repeat step 6 and 7 for the red region.

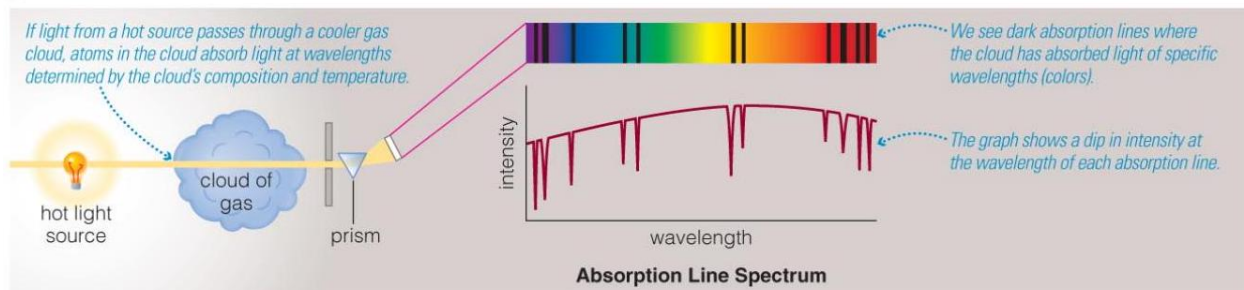
Three kinds of spectra



a



b



c

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Wavelength of Light and Spectroscopy

Name: _____

Date: _____

Partner(s): _____

Diffraction grating: _____ lines per mm

Distance between adjacent slits: $D =$ _____ nm

The angle:

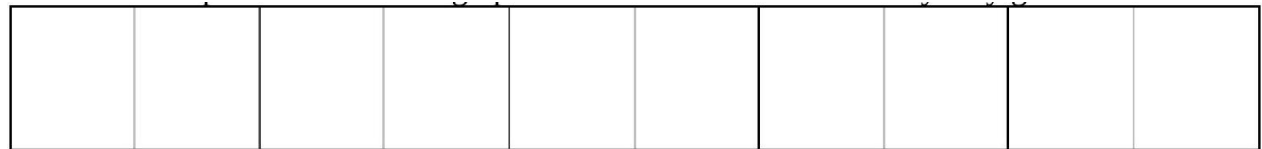
$$\theta = \frac{|\theta_{\text{right}} - \theta_{\text{left}}|}{2}$$

Wavelength: $\lambda = D \times \sin \theta$

Emission lines of mystery gas:

Color	θ_{left}	θ_{right}	θ	$\lambda(\text{nm})$

Use the colored pencils to create a graph of emission lines from the mystery gas:



300nm 350nm 400nm 450nm 500nm 550nm 600nm 650nm 700nm 750nm 800nm

Comparing this emission spectrum with the given spectra on the reference image, I infer that the gas is: _____.

Justify your answer:

Tungsten Desk Lamp

Color	θ_{left}	θ_{right}	θ	λ

What kind of spectrum is this? Circle your answer.

Continuous Emission Absorption

What's the difference between the spectrum of the lamp and the spectrum of the Sun on the reference image?

Aim

Results

Conclusion
